GENETIC ENGINEERING: GM CROPS AND CRISPR

BREEDING AND ENGINEERING TAXONOMY → BT COTTON MECHANISM AND → DMH-11 CONSTRUCT AND STATUS → CRISPR TARGETING RAIL → EDITING-OUTCOME MATRIX → SDN CLASSIFICATION LADDER

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

● PRIMARY CORE FLOW | ● SUBORDINATE EXTRA | ● APPROVAL / REVIEW

Approval: FALSE • Pending user review • active generation g1 • explicit approval of this exact generation is required • all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00

STAGE 00 BREEDING AND ENGINEERING TAXONOMY

BREEDING AND ENGINEERING TAXONOMY | CONVENTIONAL CROSS | PARENTAL GENOMES RECOMBINE SEXUALLY | OFFSPRING OF SELECTED PARENTAL LINES | SEXUALLY INCOMPATIBLE DONOR DNA | INTACT COMPATIBLE-POOL GENE | REARRANGED COMPATIBLE-POOL ELEMENTS

CONVENTIONAL CROSS → parental genomes recombine sexually

HYBRID → offspring of selected parental lines → TRANSGENIC → sexually incompatible donor DNA → CISGENIC → intact compatible-pool gene

CONCEPT / ARCHITECTURE
- CONVENTIONAL CROSS → parental genomes recombine sexually
- HYBRID → offspring of selected parental lines

MECHANISM / APPLICATION
- TRANSGENIC → sexually incompatible donor DNA
- CISGENIC → intact compatible-pool gene

READINESS / STATUS LIMIT
- INTRAGENIC → rearranged compatible-pool elements
- CONVENTIONAL CROSS → parental genomes recombine sexually

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CISGENIC → intact compatible-pool gene; INTRAGENIC → rearranged compatible-pool elements.

01

STAGE 01 BT COTTON MECHANISM AND STEWARDSHIP

BT COTTON MECHANISM AND STEWARDSHIP | CRY GENE | PLANT EXPRESSES BT PROTEIN | SPECIFIC BOLLWORM FEEDING | TOXIC ACTION IN SUSCEPTIBLE LARVAE | SELECTION PRESSURE | RESISTANT ALLELES CAN INCREASE | REFUGE + STACKING

CRY GENE → plant expresses Bt protein → SPECIFIC BOLLWORM FEEDING → toxic action in susceptible larvae → SELECTION PRESSURE → resistant alleles can increase → REFUGE + STACKING → slow resistance evolution → TRAP → insect resistance is not herbicide tolerance

STARTING CONDITION / INPUT
- CRY GENE → plant expresses Bt protein
- SPECIFIC BOLLWORM FEEDING → toxic action in susceptible larvae

SCIENTIFIC OR ENGINEERING MECHANISM
- SELECTION PRESSURE → resistant alleles can increase
- REFUGE + STACKING → slow resistance evolution

OUTPUT / FAILURE BOUNDARY
- TRAP → insect resistance is not herbicide tolerance
- CRY GENE → plant expresses Bt protein

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAP → insect resistance is not herbicide tolerance.

02

STAGE 02 DMH-11 CONSTRUCT AND STATUS

DMH-11 CONSTRUCT AND STATUS | MALE STERILITY IN ONE PARENT | FERTILITY RESTORATION IN HYBRID | SELECTABLE MARKER | ENABLE HYBRIDISATION IN MUSTARD | NO COMMERCIAL-CULTIVATION OR SETTLED-LEGALITY CLAIM

BARNASE → male sterility in one parent

BARSTAR → fertility restoration in hybrid → BAR → selectable marker → PURPOSE → enable hybridisation in mustard

CONCEPT / ARCHITECTURE
- BARNASE → male sterility in one parent
- BARSTAR → fertility restoration in hybrid

MECHANISM / APPLICATION
- BAR → selectable marker
- PURPOSE → enable hybridisation in mustard

READINESS / STATUS LIMIT
- STATUS → no commercial-cultivation or settled-legality claim
- BARNASE → male sterility in one parent

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PURPOSE → enable hybridisation in mustard; STATUS → no commercial-cultivation or settled-legality claim.

03

STAGE 03 CRISPR TARGETING RAIL

CRISPR TARGETING RAIL | GUIDE RNA | SEQUENCE RECOGNITION | TARGET DNA + PAM | PERMITTED BINDING CONTEXT | DOUBLE-STRAND CLEAVAGE | CELLULAR REPAIR | CREATES FINAL EDIT

GUIDE RNA → sequence recognition → TARGET DNA + PAM → permitted binding context → CAS9 → double-strand cleavage → CELLULAR REPAIR → creates final edit → RULE → targeting tool and repair outcome are distinct

STARTING CONDITION / INPUT

SCIENTIFIC OR ENGINEERING MECHANISM

OUTPUT / FAILURE BOUNDARY

- BARSTAR → fertility restoration in hybrid
- PURPOSE → enable hybridisation in mustard
- BARNASE → male sterility in one parent

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PURPOSE → enable hybridisation in mustard; STATUS → no commercial-cultivation or settled-legality claim.

03

STAGE 03 CRISPR TARGETING RAIL

CRISPR TARGETING RAIL | GUIDE RNA | SEQUENCE RECOGNITION | TARGET DNA + PAM | PERMITTED BINDING CONTEXT | DOUBLE-STRAND CLEAVAGE | CELLULAR REPAIR | CREATES FINAL EDIT

GUIDE RNA → sequence recognition → TARGET DNA + PAM → permitted binding context → CAS9 → double-strand cleavage → CELLULAR REPAIR → creates final edit → RULE → targeting tool and repair outcome are distinct

STARTING CONDITION / INPUT
- GUIDE RNA → sequence recognition
- TARGET DNA + PAM → permitted binding context

SCIENTIFIC OR ENGINEERING MECHANISM
- CAS9 → double-strand cleavage
- CELLULAR REPAIR → creates final edit

OUTPUT / FAILURE BOUNDARY
- RULE → targeting tool and repair outcome are distinct
- GUIDE RNA → sequence recognition

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → targeting tool and repair outcome are distinct.

04

STAGE 04 EDITING-OUTCOME MATRIX

EDITING-OUTCOME MATRIX | TEMPLATE-FREE INDELS AND KNOCKOUT | DONOR-TEMPLATE SUBSTITUTION OR INSERTION | BASE EDITING | CHEMICAL BASE CONVERSION WITHOUT DSB | PRIME EDITING | TEMPLATED SHORT WRITING WITHOUT DSB | PRECISION MECHANISMS ARE NOT INTERCHANGEABLE

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
NHEJ → template-free indels and knockout	BASE EDITING → chemical base conversion without DSB	TRAP → precision mechanisms are not interchangeable
HDR → donor-template substitution or insertion	PRIME EDITING → templated short writing without DSB	NHEJ → template-free indels and knockout

SYSTEM / DEVICE / INSTITUTION
- NHEJ → template-free indels and knockout
- HDR → donor-template substitution or insertion

DECISIVE TECHNICAL DISTINCTION
- BASE EDITING → chemical base conversion without DSB
- PRIME EDITING → templated short writing without DSB

STATUS / UNIT / EVIDENCE LIMIT
- TRAP → precision mechanisms are not interchangeable
- NHEJ → template-free indels and knockout

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PRIME EDITING → templated short writing without DSB.

05

STAGE 05 SDN CLASSIFICATION LADDER

SDN CLASSIFICATION LADDER | NO TEMPLATE; SMALL EDIT | SHORT TEMPLATE; DEFINED SMALL EDIT | GENE-SIZED TARGETED INSERTION | 2022 LIGHTER TRACK | SDN-1/2 FREE OF EXOGENOUS DNA | FULL SCRUTINY | SDN-3 REMAINS OUTSIDE EXEMPTION

SDN-1 → no template; small edit

SDN-2 → short template; defined small edit → SDN-3 → gene-sized targeted insertion → 2022 LIGHTER TRACK → SDN-1/2 free of exogenous DNA

STARTING CONDITION / INPUT
- SDN-1 → no template; small edit
- SDN-2 → short template; defined small edit

SCIENTIFIC OR ENGINEERING MECHANISM
- SDN-3 → gene-sized targeted insertion
- 2022 LIGHTER TRACK → SDN-1/2 free of exogenous DNA

OUTPUT / FAILURE BOUNDARY
- FULL SCRUTINY → SDN-3 remains outside exemption
- SDN-1 → no template; small edit

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: 2022 LIGHTER TRACK → SDN-1/2 free of exogenous DNA.

06

STAGE 06 SIX-TIER BIOSAFETY INSTITUTION MAP

SIX-TIER BIOSAFETY INSTITUTION MAP | RECOMBINANT-DNA ADVICE | INSTITUTIONAL DAY-TO-DAY OVERSIGHT | RESEARCH AND CONTAINED STAGES | ENVIRONMENTAL RELEASE AND LARGE-SCALE USE | SBCC + DLC | STATE AND DISTRICT MONITORING

RDAC → recombinant-DNA advice

IBSC → institutional day-to-day oversight → RCGM / DBT → research and contained stages → GEAC / MoEF&CC → environmental release and large-scale use

CONCEPT / ARCHITECTURE
- RDAC → recombinant-DNA advice
- IBSC → institutional day-to-day oversight

MECHANISM / APPLICATION
- RCGM / DBT → research and contained stages
- GEAC / MoEF&CC → environmental release and large-scale use

READINESS / STATUS LIMIT
- SBCC + DLC → state and district monitoring
- RDAC → recombinant-DNA advice

06

STAGE 06 SIX-TIER BIOSAFETY INSTITUTION MAP

SIX-TIER BIOSAFETY INSTITUTION MAP | RECOMBINANT-DNA ADVICE | INSTITUTIONAL DAY-TO-DAY OVERSIGHT | RESEARCH AND CONTAINED STAGES | ENVIRONMENTAL RELEASE AND LARGE-SCALE USE | SBCC + DLC | STATE AND DISTRICT MONITORING

RDAC → recombinant-DNA advice

IBSC → institutional day-to-day oversight → RCGM / DBT → research and contained stages → GEAC / MoEF&CC → environmental release and large-scale use

CONCEPT / ARCHITECTURE
- RDAC → recombinant-DNA advice
- IBSC → institutional day-to-day oversight

MECHANISM / APPLICATION
- RCGM / DBT → research and contained stages
- GEAC / MoEF&CC → environmental release and large-scale use

READINESS / STATUS LIMIT
- SBCC + DLC → state and district monitoring
- RDAC → recombinant-DNA advice

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SBCC + DLC → state and district monitoring.

07

STAGE 07 LAW AND REGULATOR SPLIT

LAW AND REGULATOR SPLIT | RULES 1989 | CORE GMO BIOSAFETY FRAMEWORK | CONTAINED WORK | ENVIRONMENTAL APPRAISAL | GM-FOOD SAFETY APPROVAL AND LABELLING TRACK | CROP SYSTEM | AGRONOMIC EVALUATION AND VARIETY PROCESSES

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
RULES 1989 → core GMO biosafety framework	GEAC → environmental appraisal	ICAR / CROP SYSTEM → agronomic evaluation and variety processes
RCGM → research / contained work	FSSAI → GM-food safety approval and labelling track	RULES 1989 → core GMO biosafety framework

SYSTEM / DEVICE / INSTITUTION
- RULES 1989 → core GMO biosafety framework
- RCGM → research / contained work

DECISIVE TECHNICAL DISTINCTION
- GEAC → environmental appraisal
- FSSAI → GM-food safety approval and labelling track

STATUS / UNIT / EVIDENCE LIMIT
- ICAR / CROP SYSTEM → agronomic evaluation and variety processes
- RULES 1989 → core GMO biosafety framework

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ICAR / CROP SYSTEM → agronomic evaluation and variety processes.

08

STAGE 08 INDIAN DECISION TIMELINE

INDIAN DECISION TIMELINE | 09 FEB 2010 | BT BRINJAL MORATORIUM | 30 MAR 2022 | SDN-1/2 EXEMPTION OM | 17 MAY 2022 | DBT GENOME-EDITED PLANT GUIDELINES | 23 JUL 2024

09 FEB 2010 → Bt brinjal moratorium → 30 MAR 2022 → SDN-1/2 exemption OM → 17 MAY 2022 → DBT genome-edited plant guidelines → 23 JUL 2024 → Supreme Court DMH-11 split verdict → 04 APR 2025 → two genome-edited rice varieties announced

CONCEPT / ARCHITECTURE
- 09 FEB 2010 → Bt brinjal moratorium
- 30 MAR 2022 → SDN-1/2 exemption OM

MECHANISM / APPLICATION
- 17 MAY 2022 → DBT genome-edited plant guidelines
- 23 JUL 2024 → Supreme Court DMH-11 split verdict

READINESS / STATUS LIMIT
- 04 APR 2025 → two genome-edited rice varieties announced
- 09 FEB 2010 → Bt brinjal moratorium

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: 23 JUL 2024 → Supreme Court DMH-11 split verdict.

09

STAGE 09 BIOSAFETY MECHANISM MAP

BIOSAFETY MECHANISM MAP | TARGET PEST PRESSURE | RESISTANCE EVOLUTION | SEED MOVEMENT | GENE-FLOW EXPOSURE | TRAIT EXPRESSION | NON-TARGET EXPOSURE PATHWAY | SIMILAR DNA SITES

TARGET PEST PRESSURE → resistance evolution

POLLEN / SEED MOVEMENT → gene-flow exposure → TRAIT EXPRESSION → non-target exposure pathway → SIMILAR DNA SITES → possible off-target edit

STARTING CONDITION / INPUT
- TARGET PEST PRESSURE → resistance evolution
- POLLEN / SEED MOVEMENT → gene-flow exposure

SCIENTIFIC OR ENGINEERING MECHANISM
- TRAIT EXPRESSION → non-target exposure pathway
- SIMILAR DNA SITES → possible off-target edit

OUTPUT / FAILURE BOUNDARY
- RESPONSE → characterisation, mitigation and monitoring
- TARGET PEST PRESSURE → resistance evolution

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SIMILAR DNA SITES → possible off-target edit; RESPONSE → characterisation, mitigation and monitoring.

TARGET PEST PRESSURE → resistance evolution

POLLEN / SEED MOVEMENT → gene-flow exposure → TRAIT EXPRESSION → non-target exposure pathway → SIMILAR DNA SITES → possible off-target edit

STARTING CONDITION / INPUT

- TARGET PEST PRESSURE → resistance evolution
- POLLEN / SEED MOVEMENT → gene-flow exposure

SCIENTIFIC OR ENGINEERING MECHANISM

- TRAIT EXPRESSION → non-target exposure pathway
- SIMILAR DNA SITES → possible off-target edit

OUTPUT / FAILURE BOUNDARY

- RESPONSE → characterisation, mitigation and monitoring
- TARGET PEST PRESSURE → resistance evolution

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SIMILAR DNA SITES → possible off-target edit; RESPONSE → characterisation, mitigation and monitoring.

10

STAGE 10

FARMER-LIVELIHOOD EVALUATION

FARMER-LIVELIHOOD EVALUATION | INTENDED AGRONOMIC CONSTRAINT | FIELD PERFORMANCE | YIELD STABILITY AND INPUT EFFECT | FARM ECONOMICS | SEED COST, LABOUR, PESTICIDE AND RISK | EXTENSION, COMPETITION, STEWARDSHIP AND MARKETS | CONTEXT-SPECIFIC WELFARE, NOT TECHNIQUE DETERMINISM

TRAIT → intended agronomic constraint

FIELD PERFORMANCE → yield stability and input effect → FARM ECONOMICS → seed cost, labour, pesticide and risk → INSTITUTIONS → extension, competition, stewardship and markets

CONCEPT / ARCHITECTURE

- TRAIT → intended agronomic constraint
- FIELD PERFORMANCE → yield stability and input effect

MECHANISM / APPLICATION

- FARM ECONOMICS → seed cost, labour, pesticide and risk
- INSTITUTIONS → extension, competition, stewardship and markets

READINESS / STATUS LIMIT

- VERDICT → context-specific welfare, not technique determinism
- TRAIT → intended agronomic constraint

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FARM ECONOMICS → seed cost, labour, pesticide and risk.

11

STAGE 11

APPROVAL AND CLAIM FIREWALL

APPROVAL AND CLAIM FIREWALL | LAB RESULT | BOUNDED MOLECULAR PROPOSITION | FIELD PERMISSION | TRIAL STAGE | FOOD APPROVAL | SEPARATE REGULATORS | VARIETY RELEASE

LAB RESULT → bounded molecular proposition → CONTAINED / FIELD PERMISSION → trial stage → ENVIRONMENTAL / FOOD APPROVAL → separate regulators → VARIETY RELEASE / SEED ACCESS / CULTIVATION → separate evidence → POST-RELEASE MONITORING → resistance, ecology and farmer outcomes

SYSTEM / DEVICE / INSTITUTION

- LAB RESULT → bounded molecular proposition
- CONTAINED / FIELD PERMISSION → trial stage

DECISIVE TECHNICAL DISTINCTION

- ENVIRONMENTAL / FOOD APPROVAL → separate regulators
- VARIETY RELEASE / SEED ACCESS / CULTIVATION → separate evidence

STATUS / UNIT / EVIDENCE LIMIT

- POST-RELEASE MONITORING → resistance, ecology and farmer outcomes
- LAB RESULT → bounded molecular proposition

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: POST-RELEASE MONITORING → resistance, ecology and farmer outcomes.

E

EXTRA

SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT | INNOVATION-SPEED DEBATE | LIGHTER TREATMENT FOR SDN-1/SDN-2 CAN ACCELERATE TRAIT DEVELOPMENT AND | BIOSAFETY DEBATE | CRITICS ARGUE THAT EVEN PRECISE EDITS MAY HAVE ECOLOGICAL | PUBLIC-TRUST DEBATE | REGULATORY DIFFERENTIATION IS SCIENTIFICALLY MEANINGFUL, BUT PUBLIC COMMUNICATION CAN | TRADE AND TRACEABILITY DEBATE

OPTIONAL SCIENTIFIC / ENGINEERING DEPTH

- CAUTION: Innovation-speed debate: lighter treatment for SDN-1/SDN-2 can accelerate trait development and public-sector crop improvement.
- CAUTION: Biosafety debate: critics argue that even precise edits may have ecological or off-target consequences needing robust oversight.
- CAUTION: Public-trust debate: regulatory differentiation is scientifically meaningful, but public communication can lag behind.
- CAUTION: Trade and traceability debate: export markets may classify or monitor edited crops differently.

READINESS, UNIT OR STATUS LIMIT

- CAUTION: Federal/agricultural politics: crop adoption interacts with farmer trust, seed systems and state-level political acceptance.
- CAUTION: Gene editing could matter for climate resilience, disease resistance, nutritional traits and public-sector breeding.
- CAUTION: GM and gene-editing capacity also reflect India's ability to convert molecular biology into agricultural technology.
- CAUTION: A nuanced regulatory model can support innovation without erasing biosafety review.

COMPARATIVE TECHNOLOGY / GOVERNANCE NUANCE

- CAUTION: Ecological uncertainty: gene flow, pollinator concerns, resistance development and biodiversity effects remain important.
- CAUTION: Institutional complexity: research oversight, biosafety testing, environmental release and agricultural release involve different bodies.
- CAUTION: Public communication gap: technical distinctions between transgenic and gene-edited products are not easy to communicate in politics.
- CAUTION: Legal uncertainty: later judicial or regulatory developments can affect rollout status, especially in contentious cases such as GM mustard.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels